

MENTAL HEALTH SENTIMENT ANALYZER

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: Mental Health Sentiment Analyzer

Technology Stack:

Python | Natural Language Processing (NLP) | Machine Learning |
Scikit-learn | TF-IDF | Streamlit

Project Description:

A Natural Language Processing based system developed to analyze employee mental health survey responses, classify sentiment, identify workplace wellbeing trends, and generate insights through an interactive Streamlit dashboard.

Submitted To

Data Vidwan

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1. Project Objective

The objective of this project is to build a Natural Language Processing system that analyzes employee mental health survey responses, classifies sentiment, identifies at-risk employee cohorts, and presents insights through a professional Streamlit dashboard.

The project supports HR teams by summarizing workplace wellness patterns from employee survey data and by providing a simple interface for analyzing new employee feedback.

2. Manager Requirement Coverage

Requirement	Status	Project File / Output
Build the model	Completed	models/sentiment_tfidf_pipeline.pkl
Develop Streamlit application with professional UI	Completed	app.py
Model should provide proper outputs	Completed with documented limitations	TF-IDF model + polarity/risk-term safeguard in Streamlit app
Prepare documentation report	Completed	reports/final_documentation_report.doc and reports/final_documentation_report.md

Include methodology, results, accuracy, achievements	Completed	This report
Prepare files for ZIP submission	Completed	Data, models, app, reports, visuals, notebooks, scripts

3. Dataset

- **Dataset:** OSMI Mental Health in Tech Survey
- **Source:** Kaggle
- **Raw records:** 1,259
- **Cleaned records:** 1,250
- **Usable free-text comment records:** 160

The dataset contains employee demographics, workplace mental health support indicators, treatment status, work interference, and optional free-text comments.

4. Work Completed

- Loaded and inspected the raw survey dataset.
- Checked missing values, duplicate records, and inconsistent values.
- Cleaned invalid age values and standardized gender categories.
- Created a cleaned survey dataset.
- Performed exploratory data analysis.
- Generated visualizations for age, gender, country, family history, treatment, work interference, sentiment distribution, model comparison, and confusion matrix.
- Preprocessed free-text employee comments.
- Generated sentiment labels using TextBlob polarity because the dataset does not contain manual sentiment labels.
- Converted text comments into TF-IDF numerical features.
- Trained Naive Bayes, Logistic Regression, Linear SVM, and tuned Logistic Regression models.
- Compared models using Accuracy and Weighted F1 score.
- Selected the best sentiment model.
- Exported the trained NLP pipeline.
- Created sentiment analysis, at-risk cohort, and HR recommendation reports.
- Developed a professional Streamlit dashboard application.

5. Methodology

5.1 Data Cleaning

The raw dataset was cleaned by removing invalid age values, standardizing gender values, filling important missing categorical fields, and keeping usable employee comments for NLP analysis.

5.2 Text Preprocessing

Employee comments were converted to lowercase, URLs were removed, special characters and numbers were removed, and extra spaces were normalized.

5.3 Sentiment Label Creation

The dataset does not provide sentiment labels. Therefore, TextBlob polarity was used to create labels:

- **Positive:** polarity greater than 0.05
- **Negative:** polarity less than -0.05
- **Neutral:** polarity between -0.05 and 0.05

5.4 Feature Extraction

TF-IDF Vectorization was used to convert cleaned text comments into numerical features for model training.

5.5 Model Training and Selection

Multiple classification models were trained and compared. The final model was selected using Weighted F1 score because it gives a better view of performance across all sentiment classes than accuracy alone.

6. Model Results

Model	Accuracy	Weighted F1
Logistic Regression	0.425	0.4197
Tuned Logistic Regression	0.425	0.4072

Naive Bayes	0.500	0.3745
Linear SVM	0.375	0.3619

7. Selected Model

The selected model is **TF-IDF Vectorizer + Logistic Regression**. It was selected because it achieved the highest Weighted F1 score among the tested sentiment models.

Exported model file: `models/sentiment_tfidf_pipeline.pkl`

8. Streamlit Application

The Streamlit application is implemented in `app.py`. It includes the following pages:

- Executive Overview
- Sentiment Analysis
- At-Risk Cohorts
- Model Performance
- Live Sentiment Prediction
- Reports Viewer

For new comment prediction, the app displays the trained model output and applies a polarity/risk-term safeguard so short high-risk phrases such as stress, burnout, anxiety, depression, and trust issues are flagged more appropriately.

9. Key Findings

- The cleaned dataset contains 1,250 survey responses.
- Only 160 rows contain usable comments, which limits the sentiment model accuracy.
- Positive comments are the largest sentiment group, but negative comments are still meaningful for wellness review.
- Employees with family history show higher treatment-seeking behavior.
- Employees reporting work interference are more likely to show mental health concerns.
- Role-based analysis could not be completed because the dataset does not contain a role/job-title column.

10. HR Recommendations

Based on the analysis of employee mental health survey responses and sentiment patterns, the following recommendations are proposed:

- Provide targeted mental health support for employees reporting frequent work interference or repeated workplace stress indicators.
- Increase awareness and accessibility of confidential mental health resources, including employee assistance programs, care options, anonymity features, and help-seeking channels.
- Introduce proactive wellness check-ins for employees showing repeated negative sentiment patterns or risk factors identified through survey analysis.
- Monitor workplace wellbeing trends across factors such as age group, gender, location, family history, and work interference to design effective wellness initiatives.
- Use sentiment analysis results as a supportive analytics tool for identifying workplace trends and improving employee wellbeing programs.
- Ensure that model predictions are not used for medical diagnosis, employee evaluation, hiring decisions, or performance management.

11. Achievements

- Completed an end-to-end NLP workflow.
- Built and exported a reusable sentiment classification pipeline.
- Created a professional Streamlit dashboard for analysis and prediction.
- Prepared final documentation and supporting reports.
- Organized datasets, models, reports, notebooks, scripts, and visuals for final ZIP submission.

12. Limitations

The model is a project-level NLP workflow and should not be treated as a medical, clinical, or employment decision system.

- Sentiment labels are generated using TextBlob, not manually annotated by domain experts.
- The free-text comment dataset is small, which affects model accuracy.
- The app is intended to support HR review, not automate final decisions.
- Role-based cohort analysis is unavailable due to missing role data.

13. Final Submission Contents

The final ZIP submission should contain:

- `app.py`

- requirements.txt
- README.md
- Data/
- models/
- reports/
- visuals/
- Notebooks/
- scripts/

14. How to Run

Install dependencies:

```
pip install -r requirements.txt
```

Run the Streamlit app:

```
python -m streamlit run app.py
```

15. Final Conclusion

The Mental Health Sentiment Analyzer project successfully developed an end-to-end NLP pipeline for analyzing employee workplace mental health feedback. The project involved data cleaning, exploratory analysis, text preprocessing, TF-IDF feature extraction, and training multiple machine learning models for sentiment classification.

Different classification approaches including Naive Bayes, Logistic Regression, and Support Vector Machine were implemented and evaluated. Based on the evaluation results, the Logistic Regression-based TF-IDF pipeline was selected as the final model and exported for integration with the Streamlit application.

The developed Streamlit dashboard provides an interactive platform for exploring employee sentiment trends, identifying potential high-risk groups, and testing new feedback inputs. The analysis highlights important workplace factors related to mental health concerns, such as work interference and treatment-seeking patterns.

Although the model performance is influenced by the limited availability of labeled text comments, the project demonstrates a complete practical workflow for applying NLP and machine learning techniques to employee wellness analytics.

The final system can support HR teams by providing data-driven insights into workplace wellbeing while emphasizing that predictions should be used as supportive indicators rather than replacements for professional assessment.